

# Bootstrapping String Theory EFT

Li-Yuan Chiang  
Yale University

with [Yu-tin Huang](#) (National Taiwan U.), [He-Chen Weng](#) (Brown U.)

“Bootstrapping string theory EFT” [2310.10710](#)

(some overlap with Justin Berman, Henriette Elvang, Aidan Herderschee [2310.10729](#))

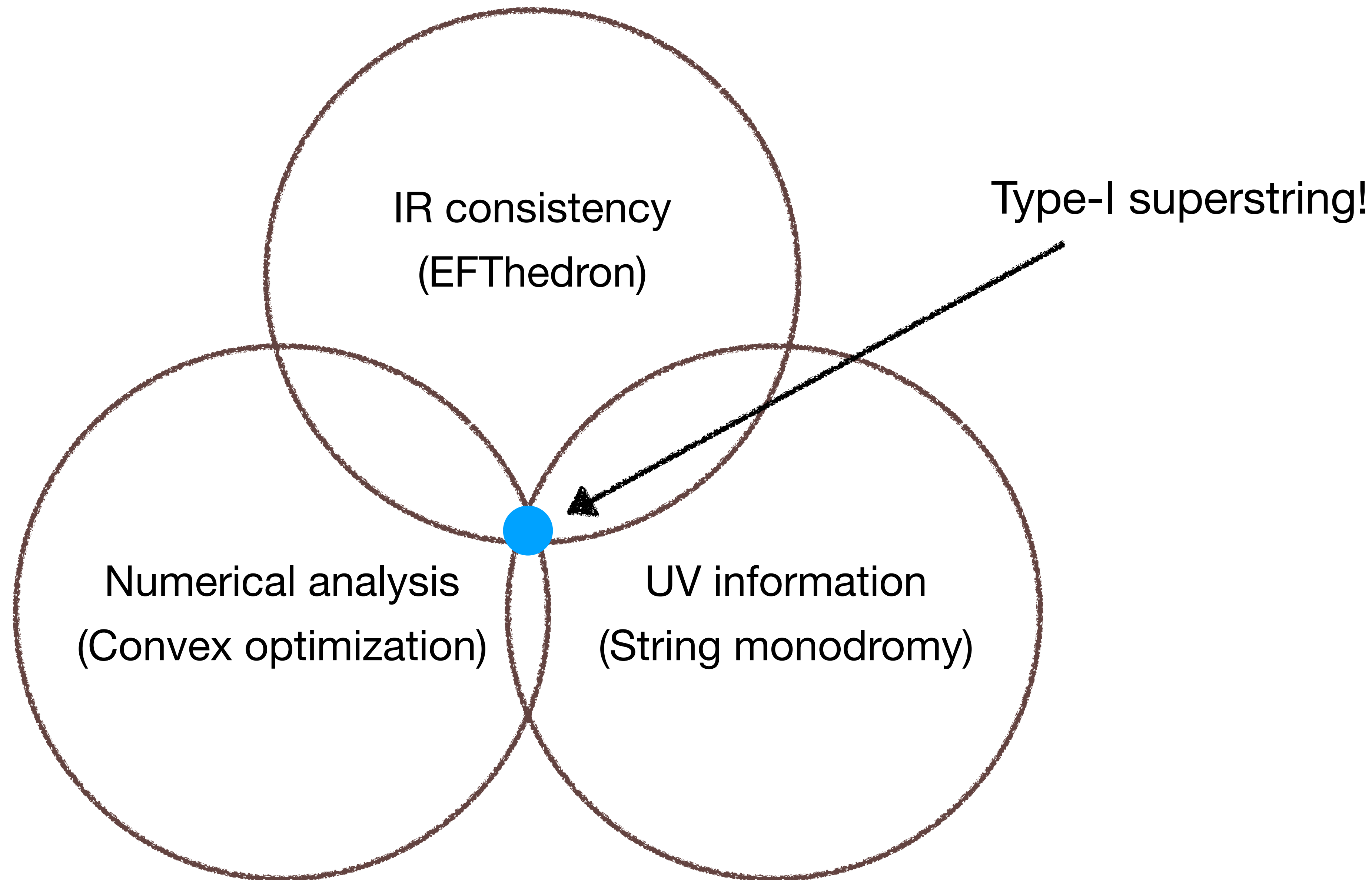


Yu-tin Huang (NTU)



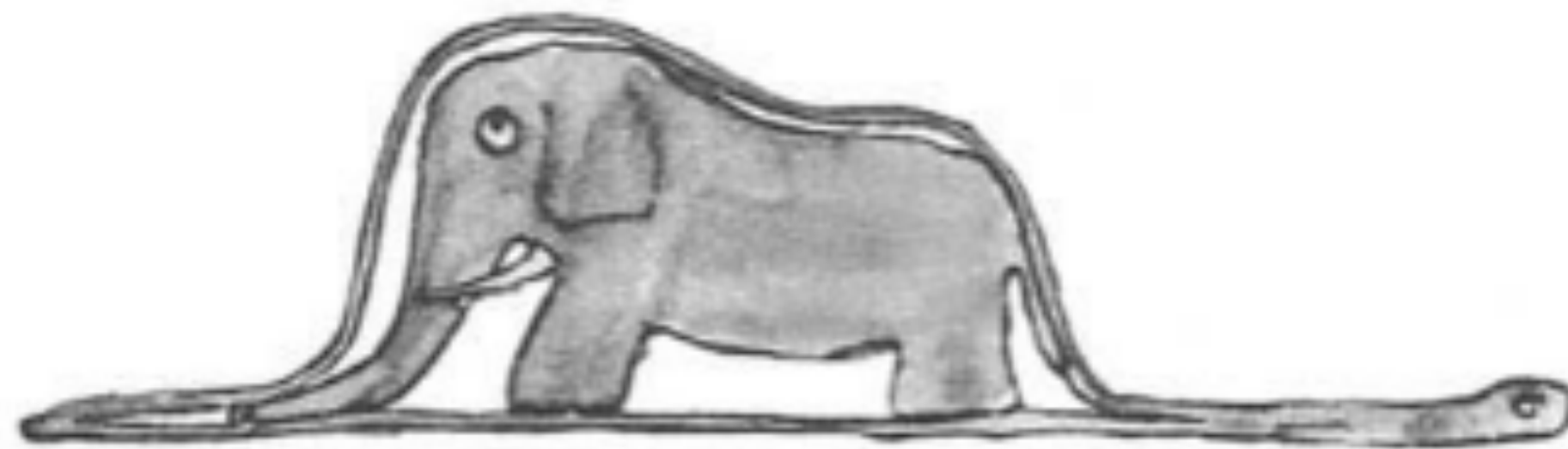
He-Chen Weng (Brown)





# EFT and IR/UV Consistency

- An **Effective Field Theory** (EFT) describes the proper degrees of freedom and their interaction below some chosen UV scale.
- An EFT in general contains an infinite number of operators compatible with the symmetries.
- Can an EFT be arbitrary?



*"With four parameters I can fit an elephant, and with five I can make him wiggle his trunk."  
— von Neumann*

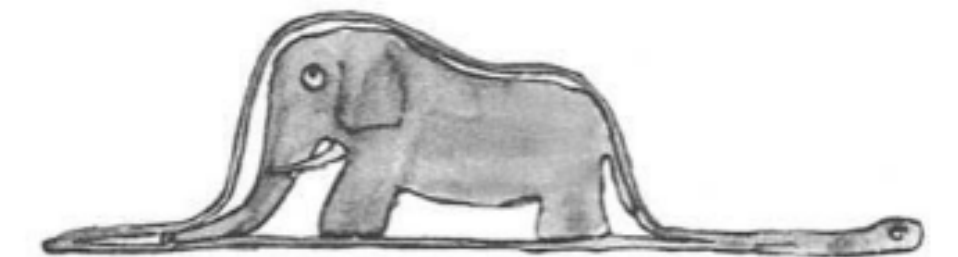
# EFT and IR/UV Consistency

- Even without the complete knowledge of the UV theory, “healthy” EFTs are subject to non-trivial constraints coming from consistency conditions.
- A famous example [1]: massless scalar with shift symmetry

$$\mathcal{L} = \partial_\mu \pi \partial^\mu \pi + \frac{c}{\Lambda^4} (\partial_\mu \pi \partial^\mu \pi)^2 + \dots$$

Consider fluctuations around  $\partial_\mu \pi = C_\mu \Rightarrow k^2 + \frac{4c}{\Lambda^4} (C \cdot k)^2 = 0$ .

We need  $c \geq 0$  for the dim-8 op. to avoid superluminal excitations!



- Lorentz invariance, unitarity, locality, and other suitable assumptions on the S-matrix  
==> **EFThedron** [2]: the allowed region in the EFT space (positive geometry)

[1] A. Adams, N. Arkani-Hamed, S. Dubovsky, A. Nicolis, R. Rattazzi [JHEP 10 \(2006\) 014](#)

[2] N. Arkani-Hamed, T.-C. Huang, Y.-T. Huang [JHEP 05 \(2021\) 259](#)

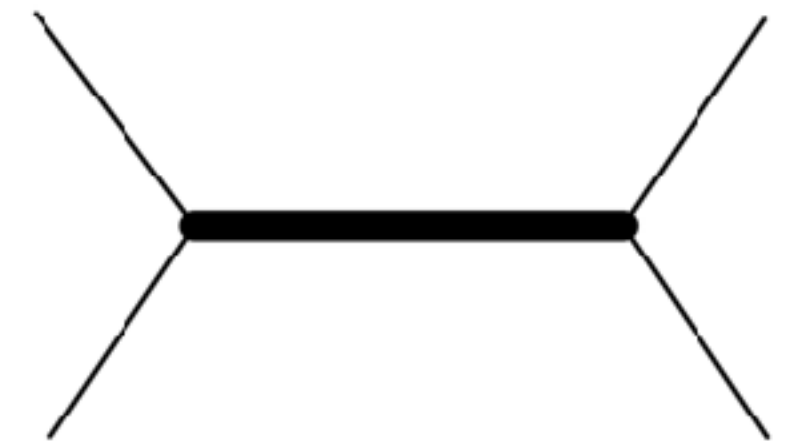


- We study consistency constraints using the EFT 4-pt amplitude, which is free from ambiguities such as gauge redundancy or field redefinition.
- Low energy expansion:

$$A(s, t) = (\text{massless poles}) + \sum_{k,q \geq 0} g_{k,q} s^{k-q} t^q \quad \longleftarrow \text{Higher derivative operators (heavy particle exchange)}$$

- Partial wave expansion:

$$A(s, t) = 16\pi \sum_l (2l+1) a_l(s) P_l \left( 1 + \frac{2t}{s} \right)$$



$$\Rightarrow 2 \operatorname{Im} a_l(s) \geq |a_l(s)|^2 \quad \Rightarrow \quad 0 \leq \operatorname{Im} a_l(s) \leq 2$$

↑  
Optical theorem (Unitarity)

↑  
Bootstrap relies on positivity!

↖  
Implications partially explored in [2]

## Assumptions:

- Analyticity at fixed small  $t$  (proven perturbatively)
- A mass gap (EFT)
- Polynomial boundedness (Froissart-Martin)

$$\lim_{|s| \rightarrow \infty} \frac{A(s, t)}{|s|^2} = 0$$

These are important assumptions that allows us to relate IR/UV physics.

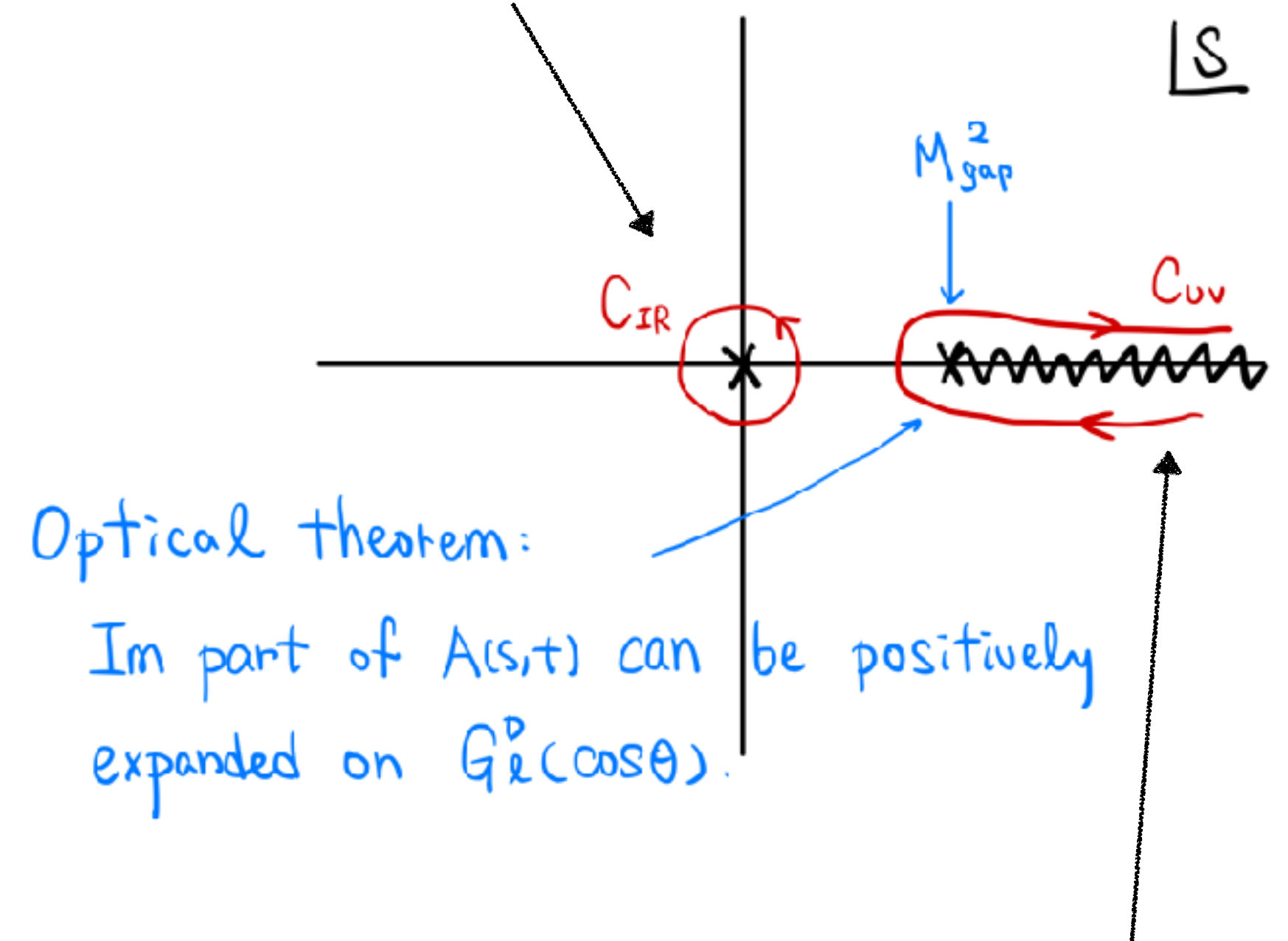
====> **Dispersive representation** (for  $k - q \geq 2$ ):

$$g_{k,q} = \frac{1}{q!} \frac{\partial^q}{\partial t^q} \int \frac{ds'}{(s')^{k-q+1}} \sum_l \rho_l(s') G_l^D \left( 1 + \frac{2t}{s'} \right)$$

**Unitarity:**  $\rho_l(s) \geq 0$

$$A(s, t) = (\text{massless poles}) + \sum_{k,q \geq 0} g_{k,q} s^{k-q} t^q$$

$$g_{k,q} = \frac{1}{2\pi i} \frac{\partial^q}{\partial t^q} \oint ds \frac{A(s, t)}{s^{k-q+1}}$$



$$\frac{1}{\pi} \int_{M^2}^{\infty} \frac{ds}{s^{k-q+1}} \text{Im } M(s, t)$$



# The EFT Bootstrap

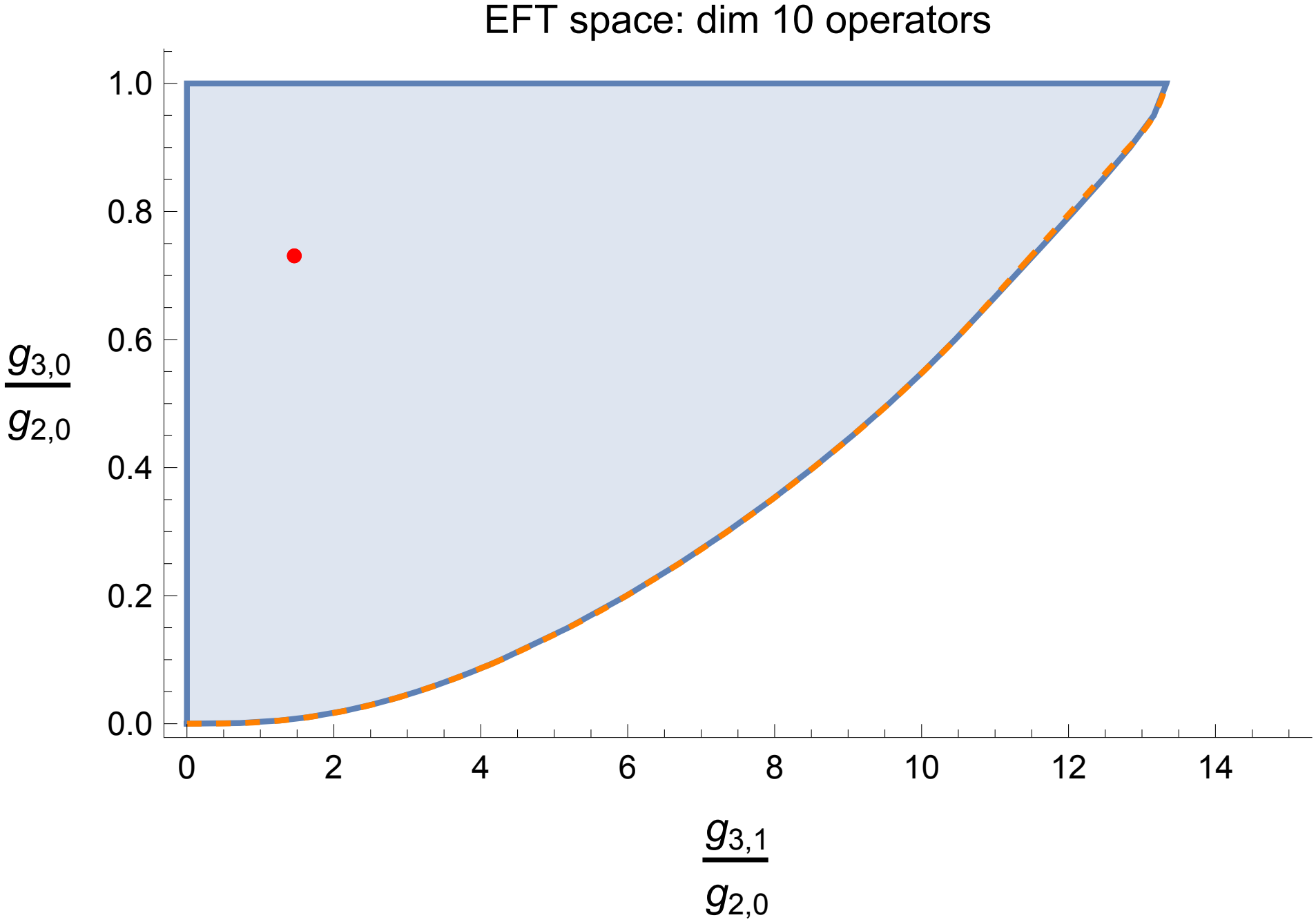
$$g_{k,q} = \int_{M_g^2}^{\infty} \frac{dm^2}{(m^2)^{k+1}} \sum_l \rho_l(m^2) v_{l,q}^D, \quad \rho_l(m^2) \geq 0$$

Polynomial of  $l(l + D - 3)$



- Positivity, together with “crossing equations” that are linear in the Wilson coefficients, enables us to carve out the EFT space using existing convex optimization methods (the *SDPB* solver that we all love)
- EFThedron geometry: the Wilson coefficients can be linearly transformed into *moment variables* and therefore satisfy an infinite tower of *Hankel constraints*, an alternative approach developed in [4].





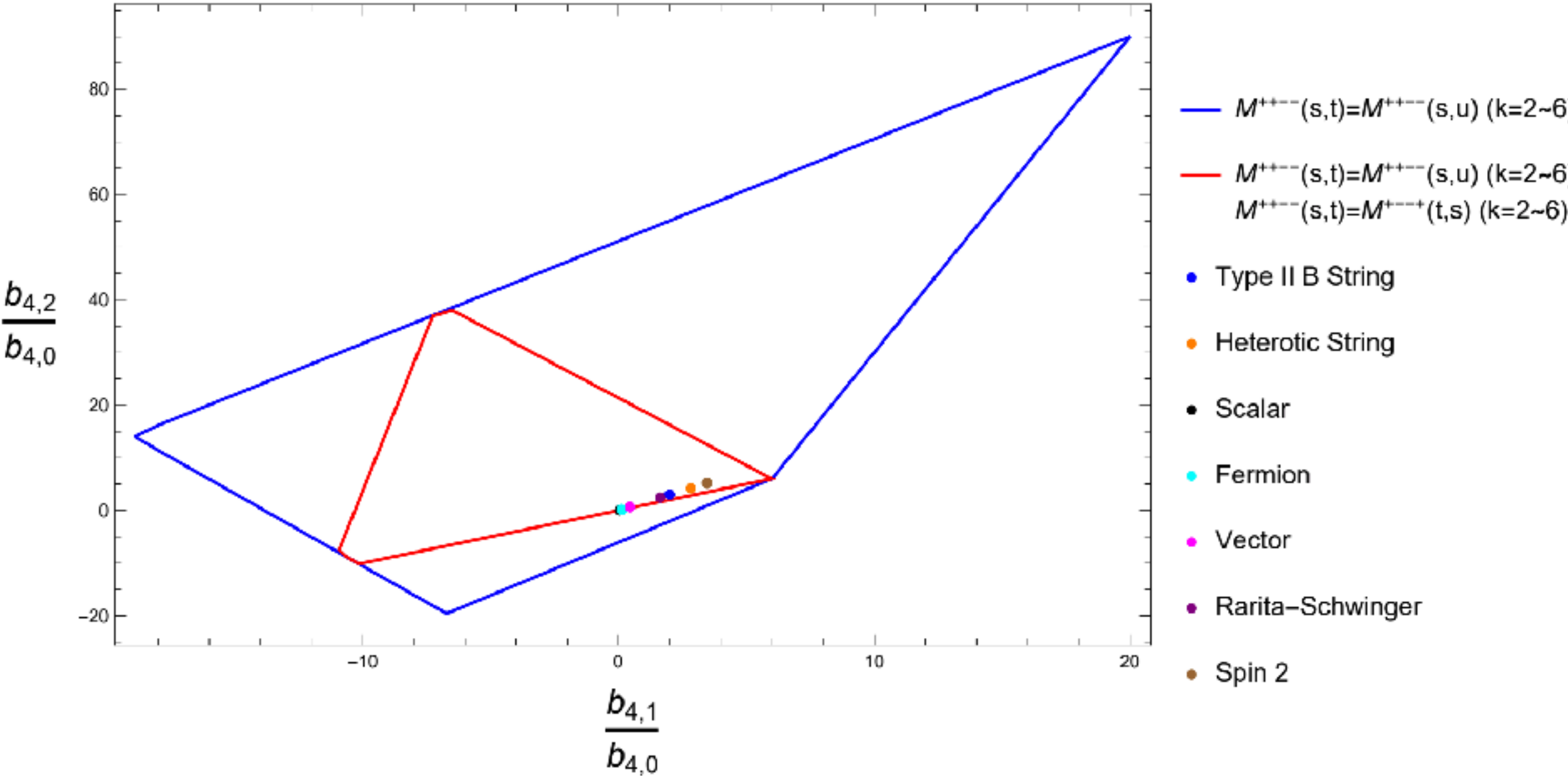
— Hankel constraints

- - - SDPB

• Type-I String

[5] Z. Bern, D. Kosmopoulos, A. Zhiboedov [2103.12728](#)

[6] L.-Y. Chiang, Y.-t. Huang, W. Li, L. Rodina, H.-C. Weng [2201.07177](#)



# String Amplitudes

- How much of the theory space can be realized by string theory?
- We need a working definition of string amplitudes: what is the signature of a world-sheet in the S-matrix?
- In [7], the authors suggested that for *open strings*, the **monodromy relation** for ordered amplitudes can serve as a universal feature.

$$A(s, u) + e^{i\pi s} A(s, t) + e^{-i\pi u} A(t, u) = 0$$

$$\begin{aligned}\frac{g_{1,0}^{max} - g_{1,0}^{min}}{g_{1,0}^{string}} &= \frac{1.20667 - 1.18890}{\zeta(3)} \approx 1.5\%, \\ \frac{g_{3,0}^{max} - g_{3,0}^{min}}{g_{3,0}^{string}} &= \frac{1.03808 - 1.03594}{\zeta(5)} \approx 0.2\%, \\ \frac{g_{4,1}^{max} - g_{4,1}^{min}}{g_{4,1}^{string}} &= \frac{0.05699 - 0.03560}{(\pi^6 - 630\zeta(3)^2)/1260} \approx 52.8\%.\end{aligned}$$

- Let us assume that the background geometry takes the form  $\mathbb{R}^{1,d-1} \times \mathbb{M}$  and the vertex operators factorize into  $\mathcal{V} = e^{ikx} V(x, k, \epsilon) U(X)$ .
- At tree level, the world-sheet has the topology of an open disk. Fixing the insertion points to  $0, z, 1, \infty$ , the scattering amplitude is given by the world-sheet integral of the CFT correlator:

$$A(s, t) \sim \int dz |z|^{2\alpha' k_1 \cdot k_2} |1 - z|^{2\alpha' k_2 \cdot k_3} f^{disk}(z, \epsilon_i, k_i)$$

K-N factor that leads to exponential softness of  $A(s, t)$  at high energy fixed angle

- We assume that  $f^{disk}$  has trivial monodromy: this ensures the presence of massless poles in each channel.



# Monodromy Relation

[8] S. Stieberger [0907.2211](#)

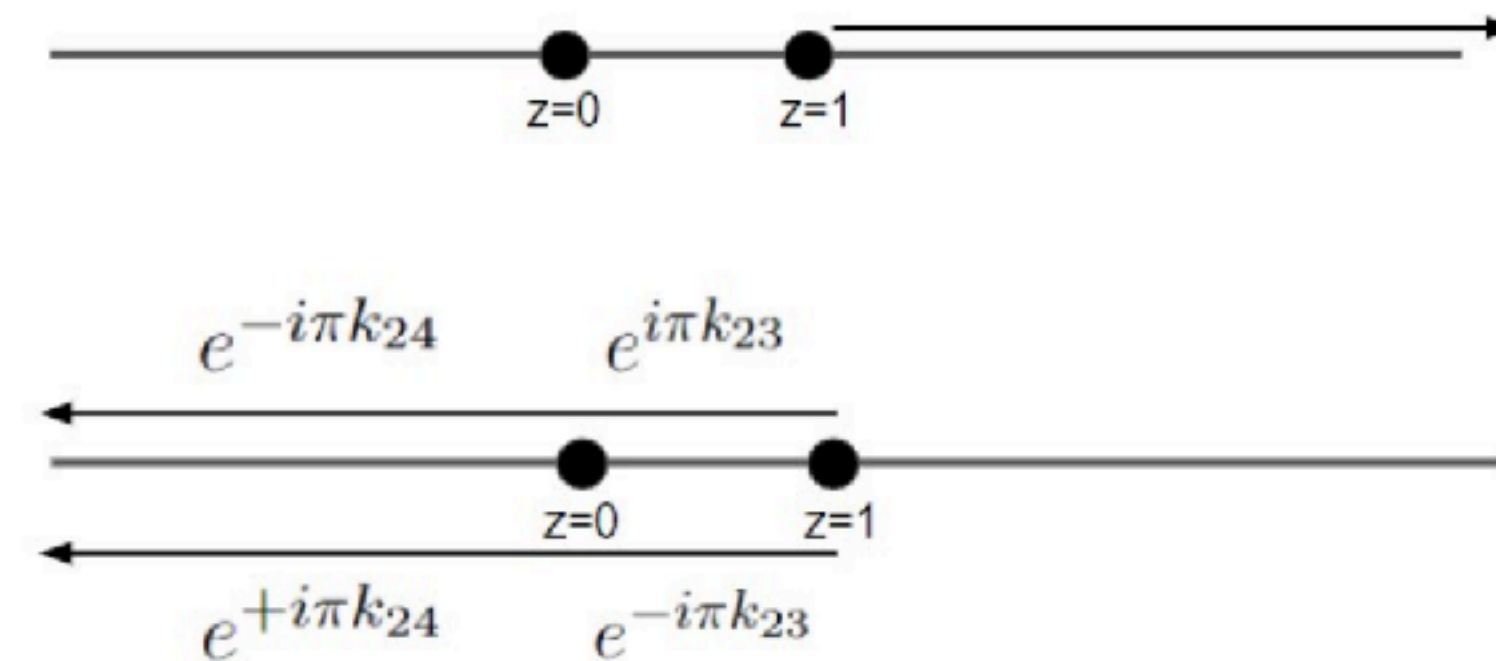
[9] N.E.J. Bjerrum-Bohr, P.H. Damgaard, P. Vanhove [Phys. Rev. Lett. 103, 161602](#)

- Amplitudes with different orderings are related by the monodromy relation [8,9].

$$A(1234) \sim \int_0^1 dz z^{k_{12}} (1-z)^{k_{23}} f^{\text{disk}}$$

$$A(1324) \sim \int_1^\infty dz z^{k_{12}} (z-1)^{k_{23}} f^{\text{disk}}$$

$$A(2134) \sim \int_{-\infty}^0 dz (-z)^{k_{12}} (1-z)^{k_{23}} f^{\text{disk}}$$



$$A(u, t) = -\mathbf{Re} \left[ e^{i\pi k_{23}} A(s, t) + e^{-i\pi k_{24}} A(s, u) \right]$$

$$0 = \mathbf{Im} \left[ e^{i\pi k_{23}} A(s, t) + e^{-i\pi k_{24}} A(s, u) \right] .$$

- The monodromy relation connects Wilson coefficients with *different* mass dimensions!  
==> powerful constraints



- Let us start with the following low-energy expansion of the amplitude that manifests maximal  $\mathcal{N} = 4$  SUSY:

$$\delta^8(Q) = \prod_{I=1}^4 \sum_{i < j} \eta_i^I \langle ij \rangle \eta_j^I$$

Pure YM
 $F^2 \phi$ 
 $F^3$

$\mathcal{A}(s, t) = \delta^8(Q) f(s, t) = \delta^8(Q) \left[ \frac{-1}{st} + b \left( \frac{1}{s} + \frac{1}{t} \right) + c \left( \frac{t}{s} + \frac{s}{t} \right) + \sum_{k, q \geq 0} \tilde{g}_{k, q} s^{k-q} t^q \right]$

- We immediately have  $b = 0$  and  $c = 0$  as they are incompatible with SUSY.
- By considering scalar scattering where  $\delta^8(Q) \sim s^2$ , we gain access to the dispersion relation for all the EFT coefficients
- The monodromy relation in the first few derivative orders:

$$\begin{pmatrix} \tilde{g}_{0,0} \\ \tilde{g}_{1,0} & \tilde{g}_{1,1} \\ \tilde{g}_{2,0} & \tilde{g}_{2,1} & \tilde{g}_{2,2} \\ \tilde{g}_{3,0} & \tilde{g}_{3,1} & \tilde{g}_{3,2} & \tilde{g}_{3,3} \\ \tilde{g}_{4,0} & \tilde{g}_{4,1} & \tilde{g}_{4,2} & \tilde{g}_{4,3} & \tilde{g}_{4,4} \end{pmatrix} = \begin{pmatrix} \frac{\pi^2}{6} & & & & \\ \tilde{g}_{1,0} & \tilde{g}_{1,1} & & & \\ \frac{\pi^4}{90} & \frac{\pi^4}{360} & & & \\ \tilde{g}_{3,0} & -\frac{\pi^2}{6} \tilde{g}_{1,0} + 2\tilde{g}_{3,0} & -\frac{\pi^2}{6} \tilde{g}_{1,0} + 2\tilde{g}_{3,0} & \tilde{g}_{3,0} & \\ \frac{\pi^6}{965} & \tilde{g}_{4,1} & -\frac{\pi^6}{15120} + 2\tilde{g}_{4,1} & \tilde{g}_{4,1} & \frac{\pi^6}{965} \end{pmatrix}$$

“Monodromy plane”

# SDPB Approach: a Lightning Review

- Bounding the Wilson coefficients involves the following optimization problem:

$\alpha$  labels the  
monodromy  
constraints

$$\text{Minimize } g_{k,q} = \sum_{\ell \in \mathcal{S}} \int_{M^2}^{\infty} ds' \rho_{\ell}(s') G_{k,q,\ell}(s') \quad \leftarrow \text{Polynomial in } s'$$

$$\text{subject to } n^{(\alpha)} = \sum_{\ell \in \mathcal{S}} \int_{M^2}^{\infty} ds' \rho_{\ell}(s') N_{\ell}^{(\alpha)}(s') = c^{(\alpha)}, \quad \leftarrow \text{"Null constraints"}$$

- We consider the dual problem

$$\text{Maximize } -\vec{c} \cdot \vec{z}$$

$$\text{subject to } \vec{z} \cdot \vec{N}_{\ell}(s') + G_{k,q,\ell}(s') \geq 0 \quad \forall s' \geq M^2 \text{ and } \ell \in \mathcal{S},$$

Mass gap

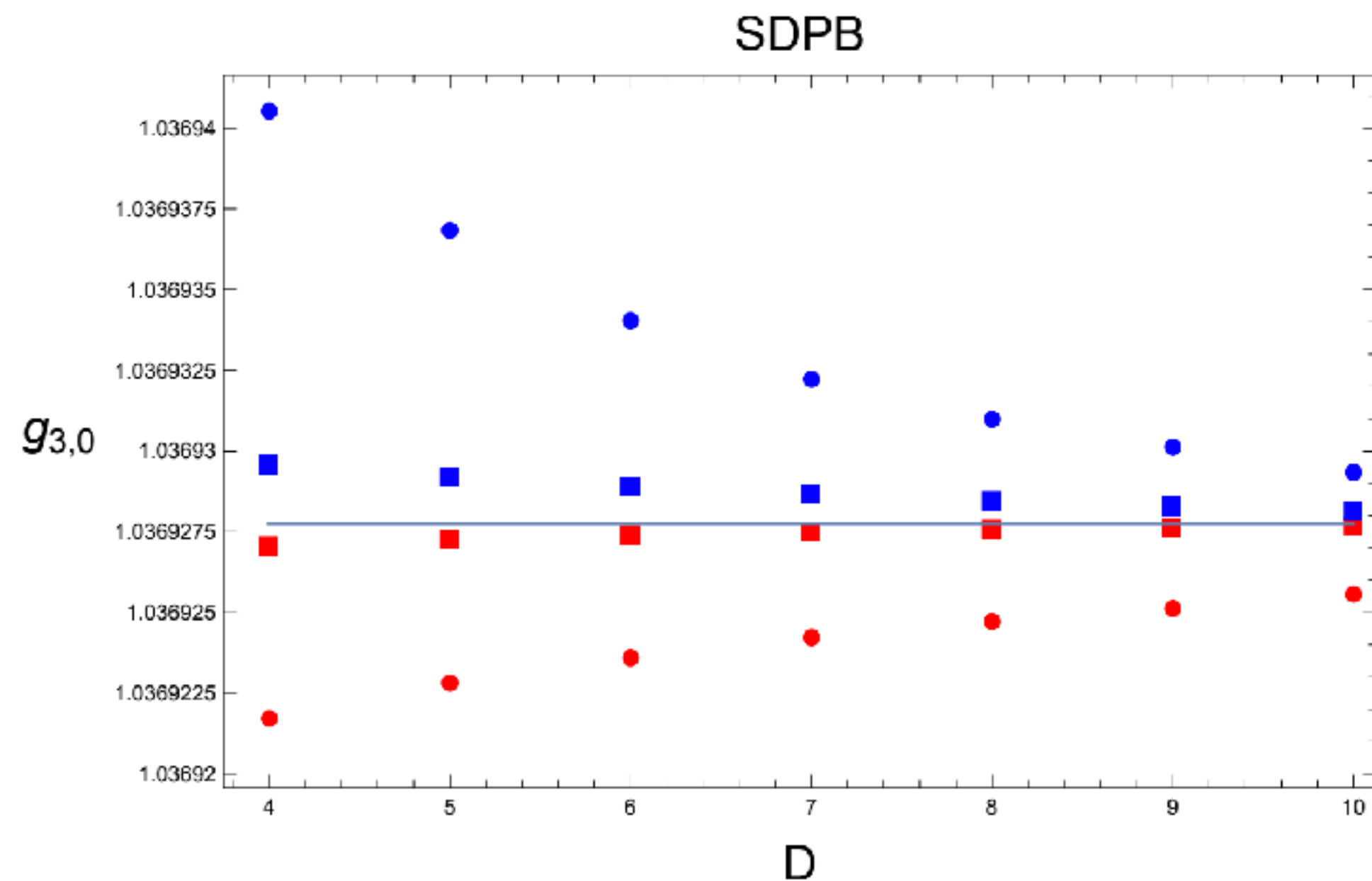
The set of allowed spins (needs a truncation  $l_{max}$ )

Valid bounds are obtained  
only when convergence with  
respect to  $l_{max}$  is established.

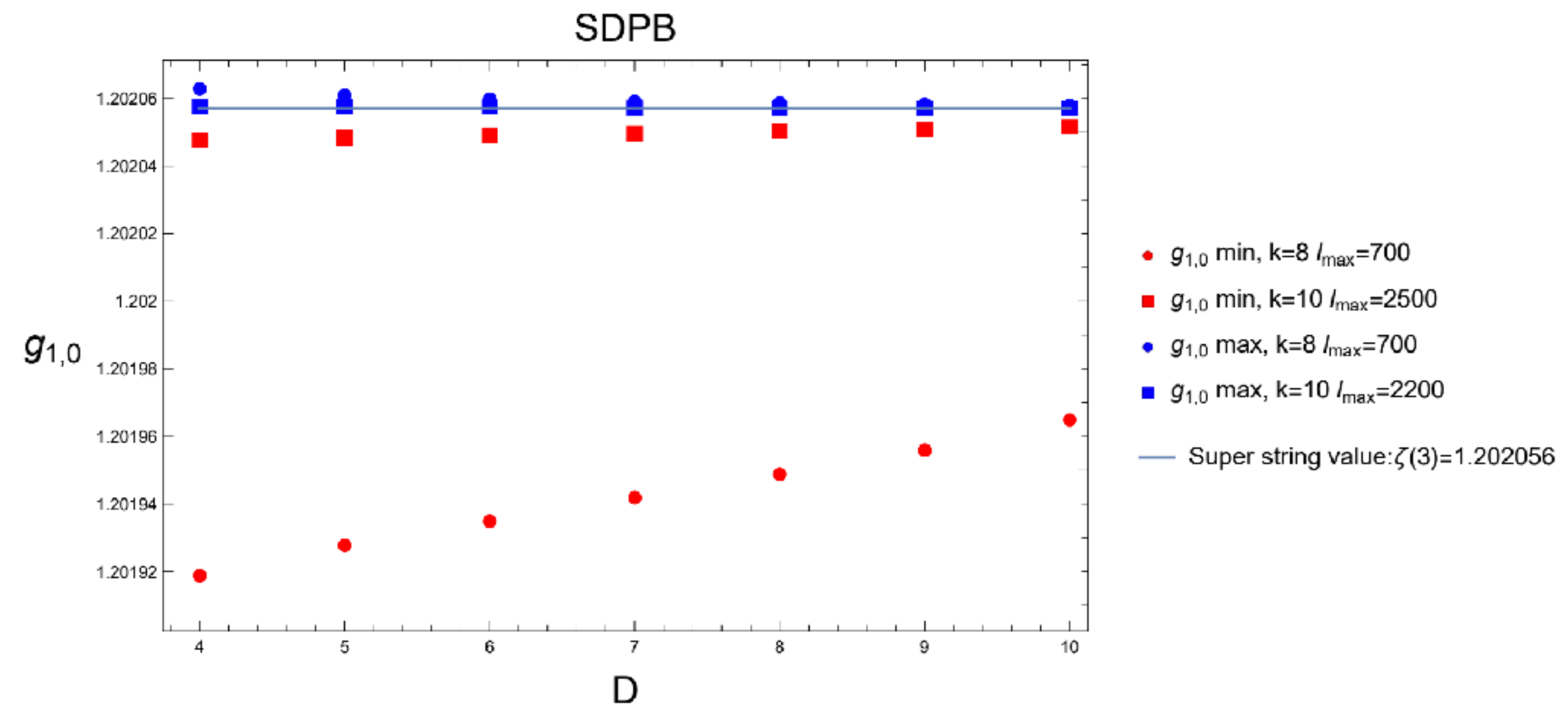
- Every dual feasible solution provides a lower bound:

$$g_{k,q} = \sum_{\ell \in \mathcal{S}} \int_{M^2}^{\infty} ds' \rho_{\ell}(s') G_{k,q,\ell}(s') \geq -\vec{z} \cdot \sum_{\ell \in \mathcal{S}} \int_{M^2}^{\infty} ds' \rho_{\ell}(s') \vec{N}_{\ell}(s') = -\vec{c} \cdot \vec{z}.$$

$$f_{string}(s, t) = \frac{\Gamma(-s) \Gamma(-t)}{\Gamma(1 - s - t)}$$



- $g_{3,0}$  min, k=8  $l_{\max}=700$
- $g_{3,0}$  min, k=10  $l_{\max}=2200$
- $g_{3,0}$  max, k=8  $l_{\max}=700$
- $g_{3,0}$  max, k=10  $l_{\max}=2200$
- Super string value:  $\zeta(5)=1.036927$



- $g_{1,0}$  min, k=8  $l_{\max}=700$
- $g_{1,0}$  min, k=10  $l_{\max}=2500$
- $g_{1,0}$  max, k=8  $l_{\max}=700$
- $g_{1,0}$  max, k=10  $l_{\max}=2200$
- Super string value:  $\zeta(3)=1.202056$

[10] J. Berman, H. Elvang, A. Herderschee [2310.10729](#)

The coefficients are isolated to a tiny island around the type-I superstring value, examined up to  $k = 8$ .

The bounds are plotted from  $D = 4$  to  $D = 10$ ,  
but there's nothing stopping us from going to  $D = 11$ .

| D  | $\tilde{g}_{k,q}$ | Two-sided bound                                 | Superstring value | Relative error       |
|----|-------------------|-------------------------------------------------|-------------------|----------------------|
| 4  | $\tilde{g}_{1,0}$ | $1.20204774 < \tilde{g}_{1,0} < 1.20205755$     | 1.20205690        | $8.1 \times 10^{-6}$ |
| 4  | $\tilde{g}_{3,0}$ | $1.03692704 < \tilde{g}_{3,0} < 1.03692956$     | 1.03692775        | $2.4 \times 10^{-6}$ |
| 4  | $\tilde{g}_{4,1}$ | $0.0405367063 < \tilde{g}_{4,1} < 0.0405469176$ | 0.0405368972      | $2.5 \times 10^{-4}$ |
| 10 | $\tilde{g}_{1,0}$ | $1.20205185 < \tilde{g}_{1,0} < 1.20205700$     | 1.20205690        | $4.3 \times 10^{-6}$ |
| 10 | $\tilde{g}_{3,0}$ | $1.03692764 < \tilde{g}_{3,0} < 1.03692814$     | 1.03692775        | $4.8 \times 10^{-7}$ |
| 10 | $\tilde{g}_{4,1}$ | $0.0405368583 < \tilde{g}_{4,1} < 0.0405426553$ | 0.0405368972      | $1.4 \times 10^{-4}$ |



# EFThedron Geometry

- The EFT coefficients can be transformed into moment variables through a linear map that *depends on the dimension D*:

$$a_{k,q} = \sum_{\ell} \int dm^2 \rho_{\ell}(m^2) \frac{J^{2q}}{m^{2(k+1)}}, \quad J^2 \equiv \ell(\ell + D - 3).$$

$$\begin{pmatrix} a_{k,0} \\ a_{k,1} \\ a_{k,2} \\ a_{k,3} \end{pmatrix} = \begin{pmatrix} 1 & 0 & 0 & 0 \\ 0 & \frac{2}{D-2} & 0 & 0 \\ 0 & \frac{-2}{D} & \frac{2}{D(D-2)} & 0 \\ 0 & \frac{8(D-1)}{3D(D+2)} & \frac{4(4-3D)}{3D(D^2-4)} & \frac{4}{3D(D^2-4)} \end{pmatrix} \begin{pmatrix} \tilde{g}_{k,0} \\ \tilde{g}_{k,1} \\ \tilde{g}_{k,2} \\ \tilde{g}_{k,3} \end{pmatrix}$$

$$\mathcal{H}_{0,0} = \sum_{\ell} \int dm^2 \rho_{\ell}(m^2) \begin{pmatrix} 1 \\ x \\ xy_{\ell} \\ x^2 \\ x^2 y_{\ell} \\ x^2 y_{\ell}^2 \\ \vdots \end{pmatrix} \begin{pmatrix} 1 \\ x \\ xy_{\ell} \\ x^2 \\ x^2 y_{\ell} \\ x^2 y_{\ell}^2 \\ \vdots \end{pmatrix}^T = \begin{pmatrix} a_{0,0} & a_{1,0} & a_{1,1} & \cdots \\ a_{1,0} & a_{2,0} & a_{2,1} & \cdots \\ a_{1,1} & a_{2,1} & a_{2,2} & \cdots \\ \vdots & \vdots & \vdots & \ddots \end{pmatrix} \geq 0$$

$$\mathcal{H}_{1,0} = \sum_{\ell} \int dm^2 \rho_{\ell}(m^2) x \begin{pmatrix} 1 \\ x \\ xy_{\ell} \\ x^2 \\ x^2 y_{\ell} \\ x^2 y_{\ell}^2 \\ \vdots \end{pmatrix} \begin{pmatrix} 1 \\ x \\ xy_{\ell} \\ x^2 \\ x^2 y_{\ell} \\ x^2 y_{\ell}^2 \\ \vdots \end{pmatrix}^T = \begin{pmatrix} a_{1,0} & a_{2,0} & a_{2,1} & \cdots \\ a_{2,0} & a_{3,0} & a_{3,1} & \cdots \\ a_{2,1} & a_{3,1} & a_{3,2} & \cdots \\ \vdots & \vdots & \vdots & \ddots \end{pmatrix} \geq 0$$

$$\mathcal{H}_{0,0} - \mathcal{H}_{1,0} \geq 0.$$

$$\mathcal{H}_{2,1} \geq 0,$$

Unitarity

Tachyon-free

Mass gap

Positive spin



(Credit: demidoffaleks)



# Critical Dimension

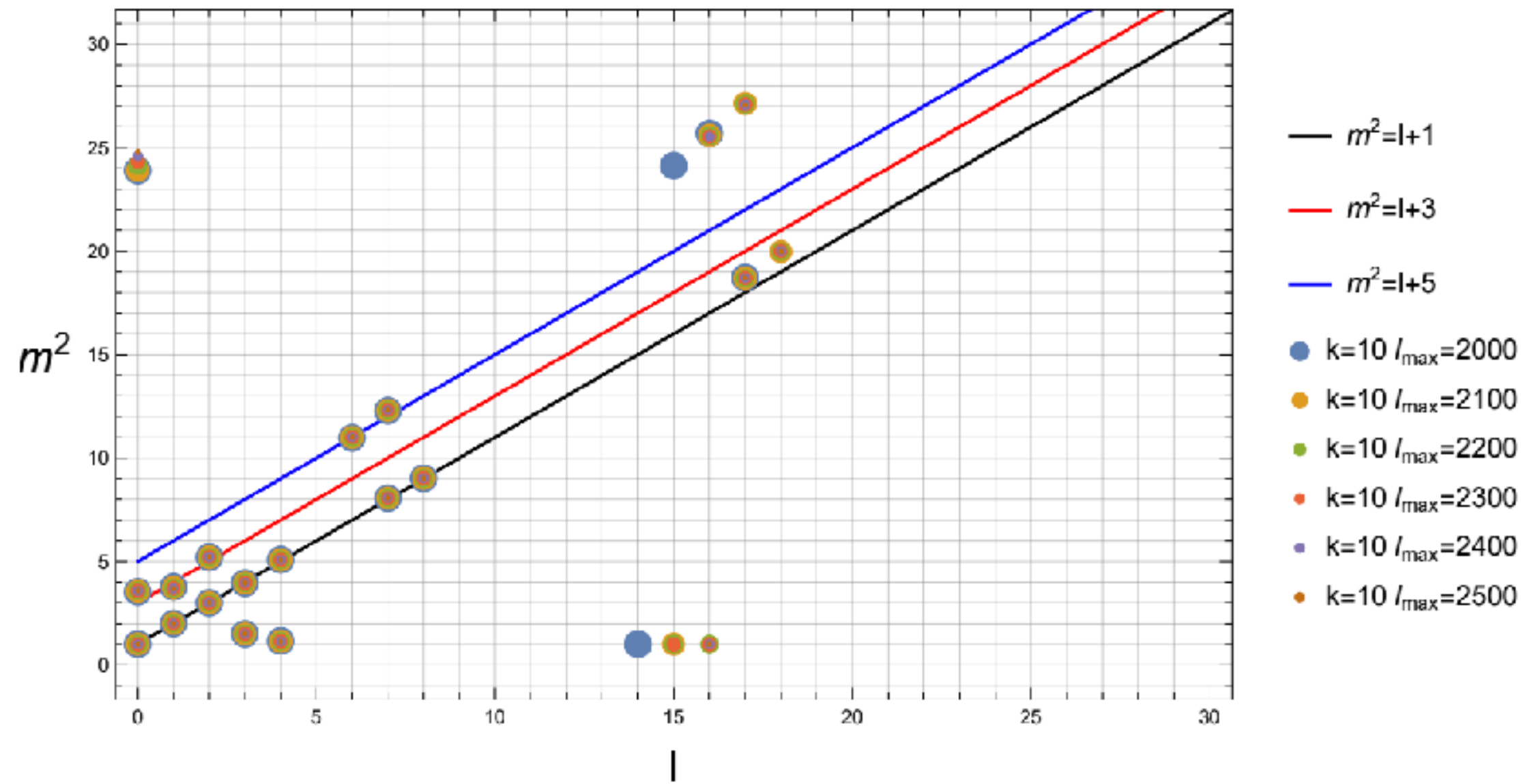
- By minimizing the violation of these positivity constraints, we obtain definite statements regarding feasibility.

|                 | $D = 10$                     | $D = 11$                     | $D = 12$                     | $D = 13$                     |
|-----------------|------------------------------|------------------------------|------------------------------|------------------------------|
| $k_{\max} = 6$  | $t^* < -1.35 \times 10^{-4}$ | $t^* < -8.45 \times 10^{-5}$ | $t^* < -7.05 \times 10^{-6}$ | $t^* < -5.51 \times 10^{-5}$ |
| $k_{\max} = 8$  | $t^* < -1.32 \times 10^{-7}$ | $t^* < -2.64 \times 10^{-7}$ | $t^* > 3.70 \times 10^{-5}$  | $t^* > 3.19 \times 10^{-4}$  |
| $k_{\max} = 10$ | $t^* < -5.53 \times 10^{-9}$ | $t^* > 8.62 \times 10^{-5}$  | $t^* > 3.43 \times 10^{-4}$  | $t^* > 1.50 \times 10^{-3}$  |

**Table 2:** Bounds on  $t^*$ , the minimal violation of Hankel matrix positivity on the monodromy plane, at different derivative orders  $k_{\max}$  and spacetime dimensions  $D$ . Bounds highlighted in red indicate a certificate of infeasibility. At  $k_{\max} = 10$ , dimensions greater than  $D_{\text{crit}} = 10$  are excluded.

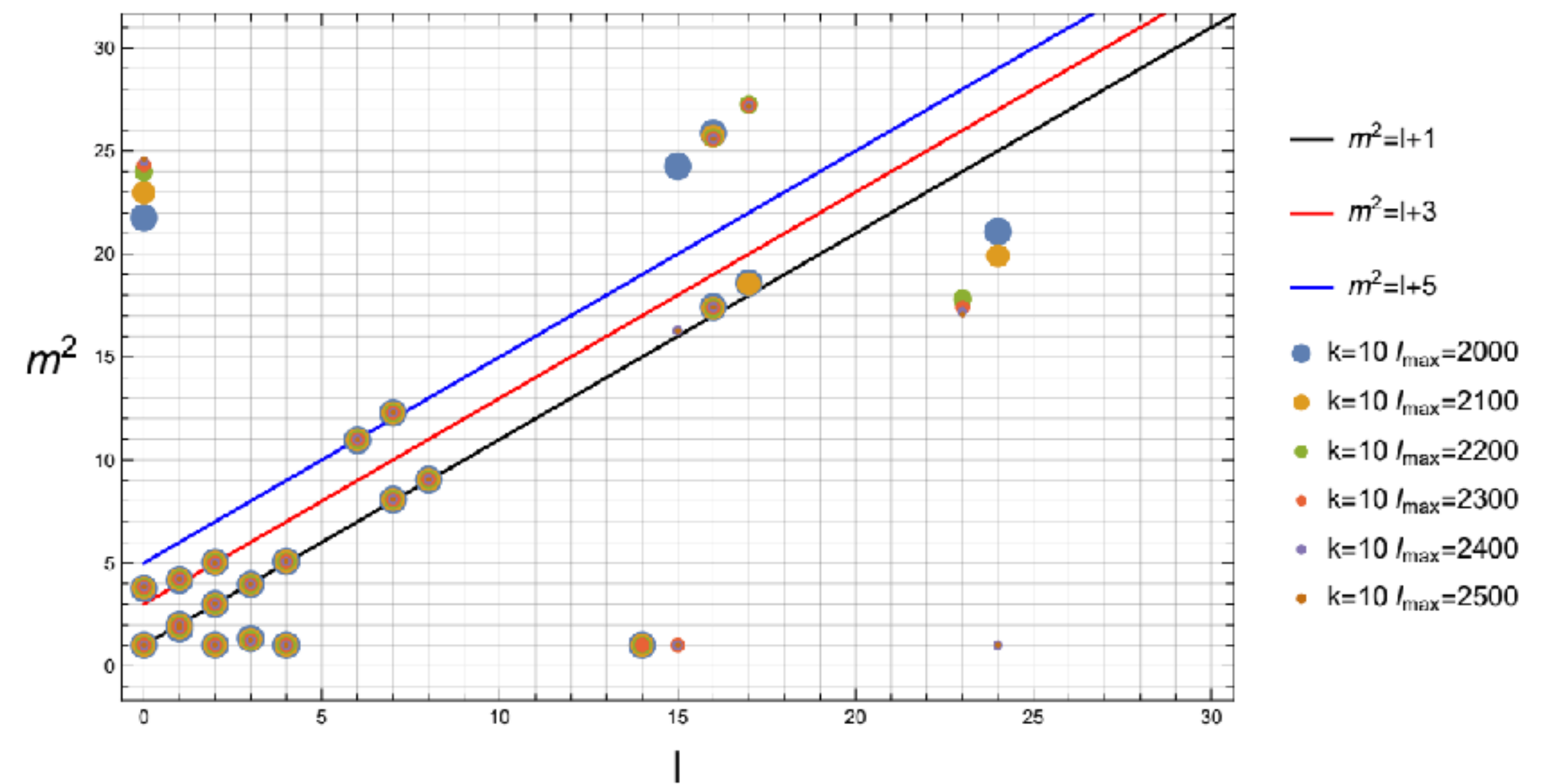
|                |                        |
|----------------|------------------------|
| beta           | 0.01                   |
| Omega_p        | $10^{10} \sim 10^{30}$ |
| Omega_d        | $10^{10} \sim 10^{30}$ |
| epsilon_gap    | $10^{-10}$             |
| epsilon_primal | $10^{-200}$            |
| epsilon_dual   | $10^{-200}$            |
| prec           | 300                    |

- While not surprising from a string theory perspective, it is remarkable that it can be established by considering only a finite number of low-energy couplings.



**Figure 3:** Extremal spectrum of  $\max g_{3,0}$  at  $k=10$ . Space-time dimension  $D=10$ .

Stable states can be found at low spin  $l \lesssim 20$  and low mass  $m^2 \lesssim 30 M_{gap}^2$



**Figure 4:** Extremal spectrum of  $\min g_{3,0}$  at  $k=10$ . Space-time dimension  $D=10$ .

# Relaxing SUSY

- non-SUSY scalar EFT

$$A(s, t) = -\left(\frac{s}{t} + \frac{t}{s}\right) + b\left(\frac{1}{s} + \frac{1}{t}\right) + \sum_{k,q \geq 0} g_{k,q} s^{k-q} t^q.$$

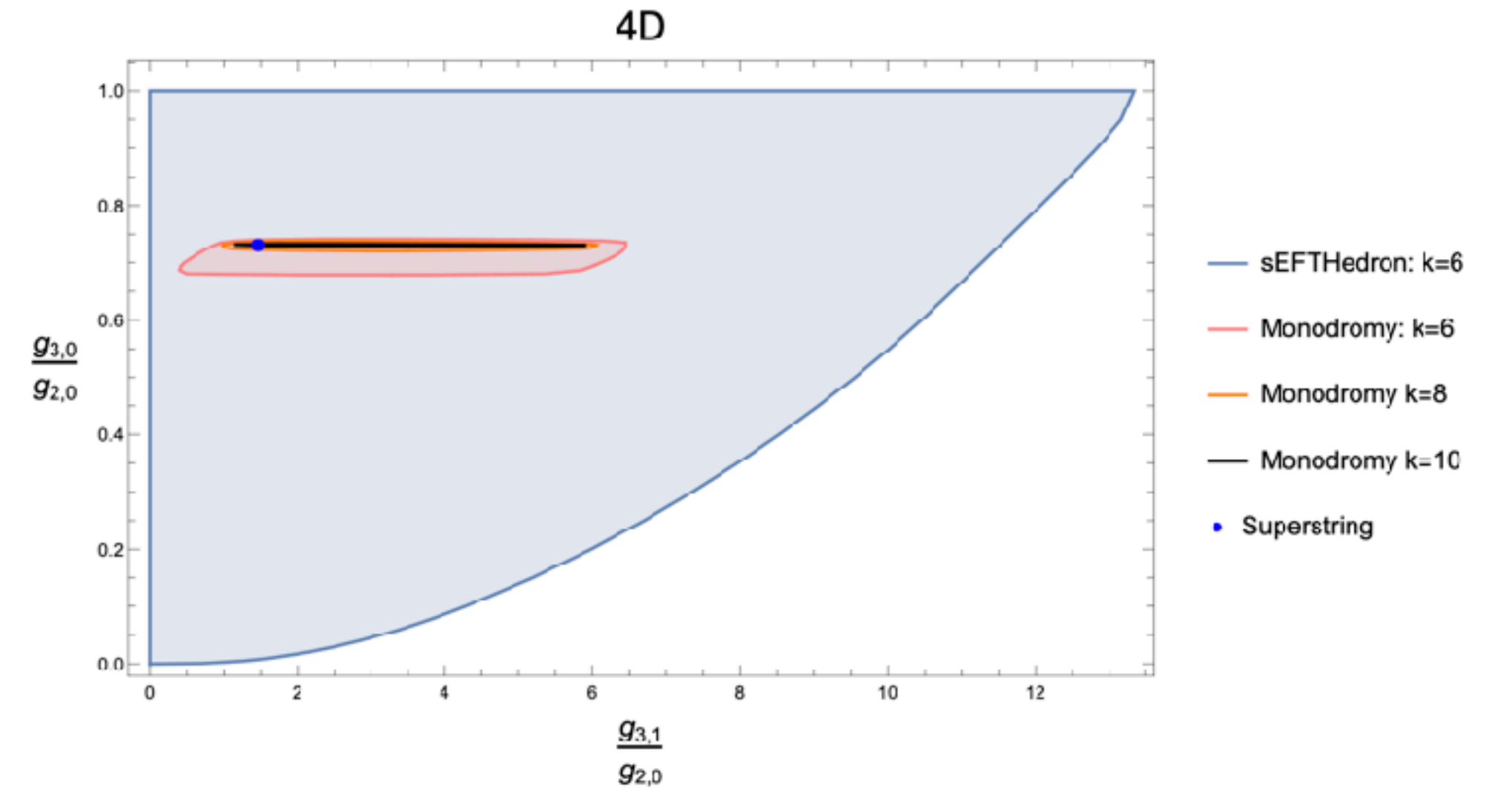
$$\begin{pmatrix} g_{0,0} \\ g_{1,0} & g_{1,1} \\ g_{2,0} & g_{2,1} & g_{2,2} \\ g_{3,0} & g_{3,1} & g_{3,2} & g_{3,3} \end{pmatrix} = \begin{pmatrix} -1 & & & \\ g_{1,0} & g_{1,0} & & \\ \frac{\pi^2}{6} & \frac{\pi^2}{6} & \frac{\pi^2}{6} & \\ g_{3,0} & 2g_{3,0} - \frac{\pi^2}{6}g_{1,0} & 2g_{3,0} - \frac{\pi^2}{6}g_{1,0} & g_{3,0} \end{pmatrix}$$

| D  | $g_{k,q}$ | Two-sided bound               | Superstring value | Relative error       |
|----|-----------|-------------------------------|-------------------|----------------------|
| 4  | $g_{3,0}$ | $1.2012 < g_{3,0} < 1.20369$  | 1.202056          | $2.0 \times 10^{-3}$ |
| 4  | $g_{3,1}$ | $1.91 < g_{3,1} < 9.8125$     | 2.404             | 3.2                  |
| 10 | $g_{3,0}$ | $1.20186 < g_{3,0} < 1.20266$ | 1.202056          | $6.6 \times 10^{-4}$ |
| 10 | $g_{3,1}$ | $1.99 < g_{3,1} < 5.09$       | 2.404             | 1.2                  |

$$k_{\max} = 12, l_{\max} = 1600$$

|                 | $D = 12$                       | $D = 13$                      | $D = 14$                     | $D = 15$                     |
|-----------------|--------------------------------|-------------------------------|------------------------------|------------------------------|
| $k_{\max} = 20$ | ...                            | ...                           | $t^* = 2.81 \times 10^{-15}$ | $t^* = 7.37 \times 10^{-10}$ |
| $k_{\max} = 22$ | ...                            | $t^* = -7.62 \times 10^{-10}$ | $t^* = 1.64 \times 10^{-10}$ | ...                          |
| $k_{\max} = 26$ | $t^* = -8.998 \times 10^{-25}$ | $t^* = 2.276 \times 10^{-14}$ | ...                          | ...                          |

**Table 3:** Minimal violation of unitarity in various spacetime dimensions. The current best bound is  $D \leq 12$ , obtained at derivative order  $k_{\max} = 26$ .



**Figure 7:** Comparing the allowed region of  $\left(\frac{g_{3,0}}{g_{2,0}}, \frac{g_{3,1}}{g_{2,0}}\right)$  by imposing crossing relations alone (sEFTHedron [19]) and imposing monodromy relations (up to  $k=10$ ). The bounds on coupling  $g_{3,0}$  shrink toward superstring value  $\zeta(3)$ , however, the bounds on  $g_{3,1}$  remain finite.



- Let us now consider the gluon EFT in 4D

$$A(1^-2^+3^-4^+) = \langle 13 \rangle^2 [24]^2 \left( \frac{-1}{st} + \sum_{k,q} b_{k,q} s^{k-q} t^q \right)$$

$$A(1^-2^-3^+4^+) = \langle 12 \rangle^2 [34]^2 \left( \frac{-1}{st} + \frac{\tilde{a}}{s} + \tilde{c} \frac{t}{s} + \sum_{k,q} \tilde{b}_{k,q} s^{k-q} t^q \right)$$

$$A(1^-2^+3^-4^+) = -u^2 \sum_l \int dm^2 \frac{d_{-2,-2}^{\ell \geq 2}(\theta)}{\cos^4 \frac{\theta}{2} m^4} \frac{\rho_\ell^{+-}(m^2)}{s - m^2}$$

$$A(1^-2^-3^+4^+) = -s^2 \sum_l \int dm^2 \frac{d_{0,0}^{\ell \geq 0}(\theta)}{m^4} \frac{\rho_\ell^{++}(m^2)}{s - m^2}$$

$$A(1^-3^+2^-4^+) = -\mathbf{Re} \left[ e^{i\pi(k_2 \cdot k_3)} A(1^-2^-3^+4^+) + e^{i\pi(-k_2 \cdot k_4)} A(2^-1^-3^+4^+) \right]$$

$$0 = \mathbf{Im} \left[ e^{i\pi(k_2 \cdot k_3)} A(1^-2^-3^+4^+) + e^{i\pi(-k_2 \cdot k_4)} A(2^-1^-3^+4^+) \right]$$

$$\begin{pmatrix} b_{0,0} \\ b_{1,0} & b_{1,1} \\ b_{2,0} & b_{2,1} & b_{2,2} \\ b_{3,0} & b_{3,1} & b_{3,2} & b_{3,3} \end{pmatrix} = \begin{pmatrix} -\tilde{c} + \frac{\pi^2}{6} & & & \\ \tilde{b}_{1,0} & \tilde{b}_{1,0} & & \\ \frac{\pi^4}{45} - \tilde{b}_{2,0} & \frac{\tilde{c}\pi^2}{6} + \frac{\pi^4}{40} - 2\tilde{b}_{2,0} & \frac{\pi^4}{45} - \tilde{b}_{2,0} & \\ \tilde{b}_{3,0} & -\frac{\pi^2}{4}\tilde{b}_{1,0} + 3\tilde{b}_{3,0} - \frac{1}{2}\tilde{b}_{3,2} & -\frac{\pi^2}{4}\tilde{b}_{1,0} + 3\tilde{b}_{3,0} - \frac{1}{2}\tilde{b}_{3,2} & \tilde{b}_{3,0} \end{pmatrix}$$

$$\begin{pmatrix} \tilde{b}_{0,0} \\ \tilde{b}_{1,0} & \tilde{b}_{1,1} \\ \tilde{b}_{2,0} & \tilde{b}_{2,1} & \tilde{b}_{2,2} \\ \tilde{b}_{3,0} & \tilde{b}_{3,1} & \tilde{b}_{3,2} & \tilde{b}_{3,3} \end{pmatrix} = \begin{pmatrix} \tilde{c} + \frac{\pi^2}{6} & & & \\ \tilde{b}_{1,0} & \tilde{b}_{1,0} & & \\ \tilde{b}_{2,0} & -\frac{\tilde{c}\pi^2}{6} - \frac{\pi^4}{120} + \tilde{b}_{2,0} & -\frac{\tilde{c}\pi^2}{6} + \frac{\pi^4}{90} & \\ \tilde{b}_{3,0} & -\frac{\pi^2}{12}\tilde{b}_{1,0} + \tilde{b}_{3,0} + \frac{1}{2}\tilde{b}_{3,2} & \tilde{b}_{3,2} & \frac{\pi^2}{12}\tilde{b}_{1,0} + \frac{1}{2}\tilde{b}_{3,2} \end{pmatrix}$$

| Wilson coefficients | Two-sided bound                                    | Superstring value | Relative error       |
|---------------------|----------------------------------------------------|-------------------|----------------------|
| $\tilde{b}_{1,0}$   | $1.20203 < \tilde{b}_{1,0} < 1.202059$             | 1.202056          | $2.4 \times 10^{-5}$ |
| $\tilde{b}_{2,0}$   | $1.08231 < \tilde{b}_{2,0} < 1.08233$              | 1.082323          | $1.8 \times 10^{-5}$ |
| $\tilde{b}_{3,2}$   | $0.09653 < \tilde{b}_{3,2} < 0.0966$               | 0.09655           | $7.2 \times 10^{-4}$ |
| $\tilde{c}$         | $-1 \times 10^{-6} < \tilde{c} < 2 \times 10^{-5}$ | 0                 | N/A                  |

Coefficients are pinned down to the superstring values even *without* assuming SUSY!

$$k_{max} = 10, l_{max} = 1500$$



- To deviate from the superstring value while maintaining unitarity, we introduce tachyons:

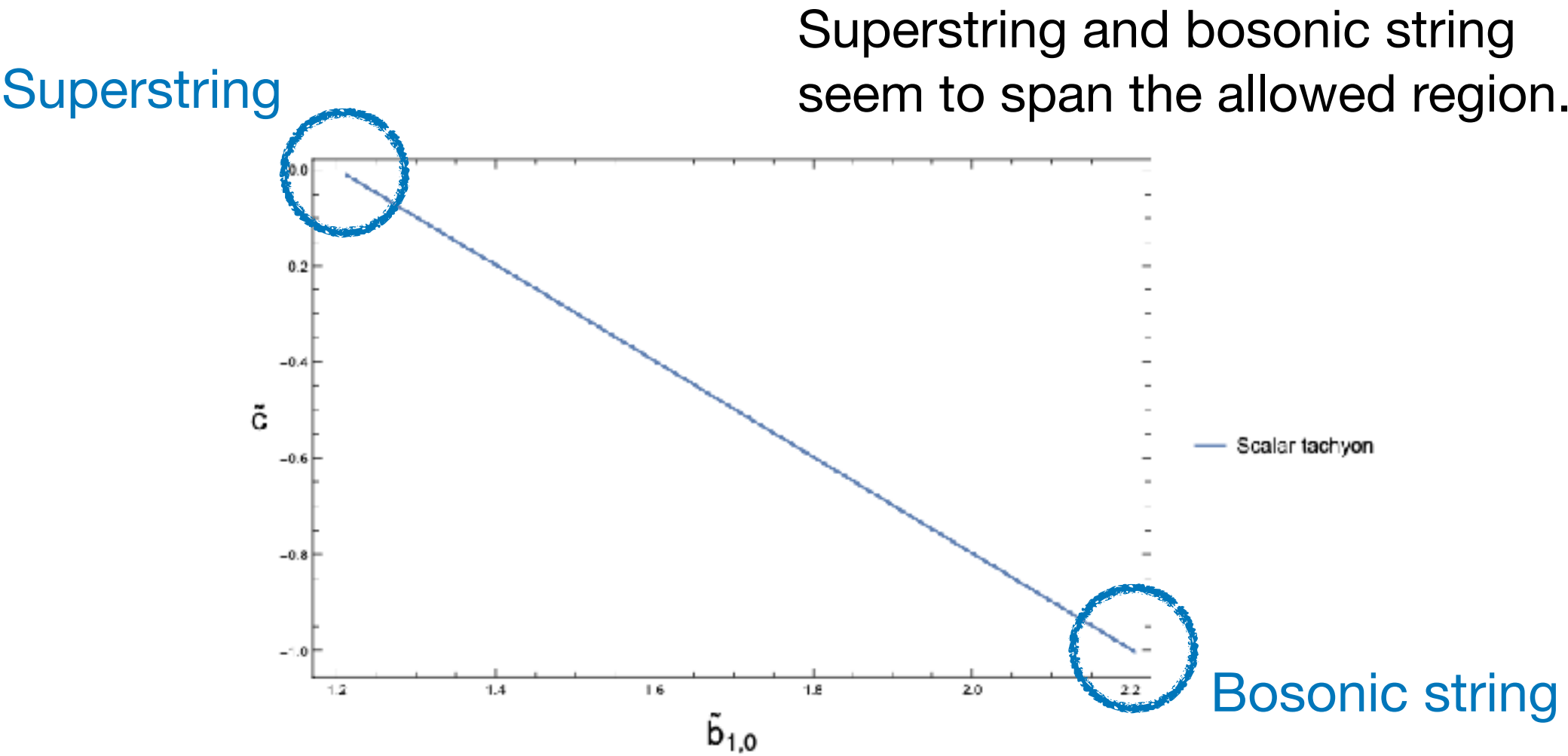
$$A(1^-2^+3^-4^+) = -u^2 \sum_l \int dm^2 \frac{d_{-2,-2}^{\ell \geq 2}(\theta)}{\cos^4 \frac{\theta}{2} m^4} \frac{|p_\ell(m^2)^{+-}|^2}{s-m^2}$$

$$A(1^-2^-3^+4^+) = -s^2 \sum_l \int dm^2 \left( \delta(m^2+M^2) \frac{d_{0,0}^{\ell \geq 0}(\theta)}{m^4} \frac{|p_\ell^{++}|^2}{s-m^2} + \frac{d_{0,0}^{\ell \geq 0}(\theta)}{m^4} \frac{|p_\ell(m^2)^{++}|^2}{s-m^2} \right)$$

Tachyon

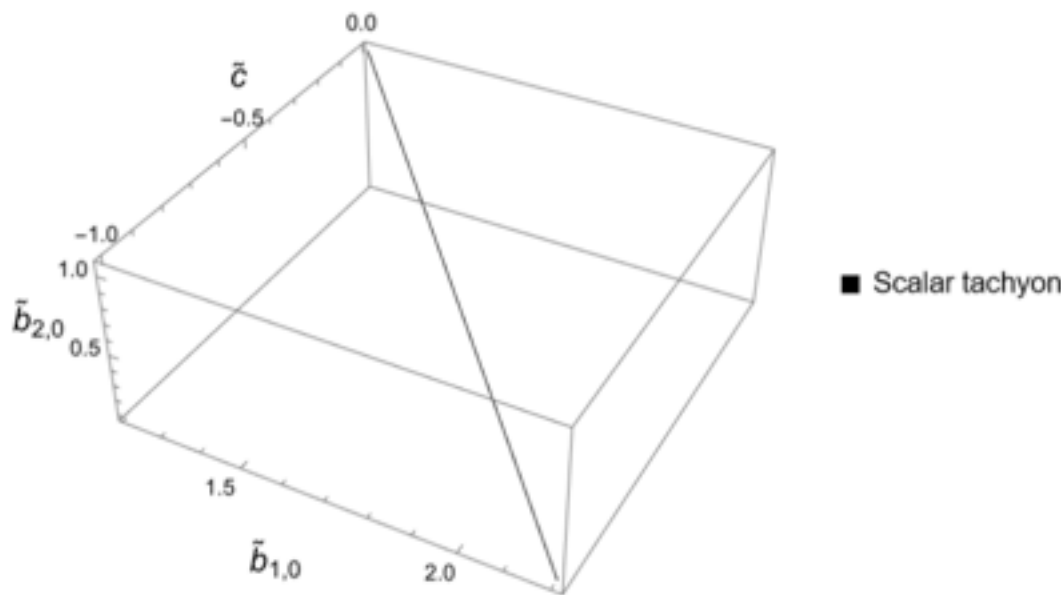
| Wilson coefficients | Arbitrary spin tachyon bound                | Bosonic string value |
|---------------------|---------------------------------------------|----------------------|
| $\tilde{b}_{1,0}$   | $1.20193 < \tilde{b}_{1,0} < 2.208$         | 2.20205              |
| $\tilde{b}_{2,0}$   | $0.07817 < \tilde{b}_{2,0} < 1.0827$        | 0.08232              |
| $\tilde{b}_{3,2}$   | $0.0959 < \tilde{b}_{3,2} < 0.8592$         | 0.85573              |
| $\tilde{c}$         | $-1.00418 < \tilde{c} < 3.3 \times 10^{-4}$ | -1                   |
| Wilson coefficients | Scalar tachyon bound                        | Bosonic string value |
| $\tilde{b}_{1,0}$   | $1.20203 < \tilde{b}_{1,0} < 2.204$         | 2.20205              |
| $\tilde{b}_{2,0}$   | $0.0802 < \tilde{b}_{2,0} < 1.08233$        | 0.08232              |
| $\tilde{b}_{3,2}$   | $0.0964 < \tilde{b}_{3,2} < 0.8573$         | 0.85573              |
| $\tilde{c}$         | $-1.002 < \tilde{c} < 2 \times 10^{-5}$     | -1                   |

Allowing arbitrary spin for the tachyon does not lead to significant change in the bounds.



**Figure 9:** The allowed region of  $(\tilde{b}_{1,0}, \tilde{c})$ . This is a two-dimensional region but the width of the region is small. For most fixed  $\tilde{b}_{1,0}$ , the discrepancy between of the bound of  $\tilde{c}$  is  $\left| \frac{\tilde{c}_{max} - \tilde{c}_{min}}{\tilde{c}_{max}} \right| \sim O(10^{-3})$ .

| Error of bosonic string    | Error of Superstring       |
|----------------------------|----------------------------|
| (Max) $8.8 \times 10^{-4}$ | (Min) $1.6 \times 10^{-5}$ |
| (Min) $2.5 \times 10^{-2}$ | (Max) $9.2 \times 10^{-6}$ |
| (Max) $1.8 \times 10^{-3}$ | (Min) $1.5 \times 10^{-3}$ |
| (Min) $2 \times 10^{-3}$   | (Max) N/A                  |



- Allowed region for closed string EFT can be obtained by implementing the KLT relation [11]

$$M(s, t) = 8G \sin(\pi t) A(s, t) A(t, u)$$

- While crossing symmetry is manifest due to the monodromy relation, unitarity is not guaranteed and requires further checking.

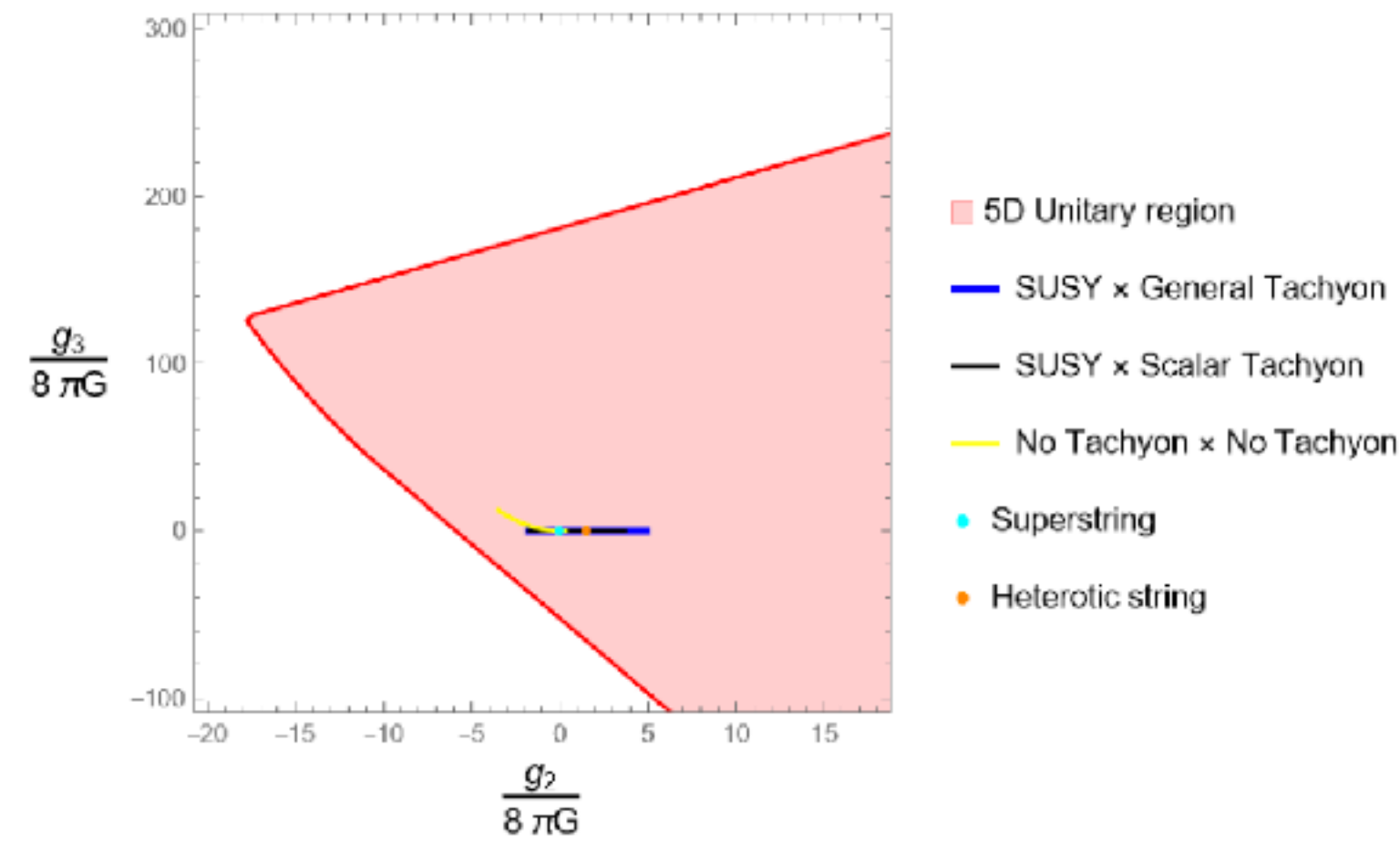
- The four-scalar amplitude can be parametrized as

$$A(s, t) = -\left(\frac{s}{t} + \frac{t}{s}\right) + b\left(\frac{1}{s} + \frac{1}{t}\right) + \sum_{k, q \geq 0} g_{k, q} s^{k-q} t^q.$$

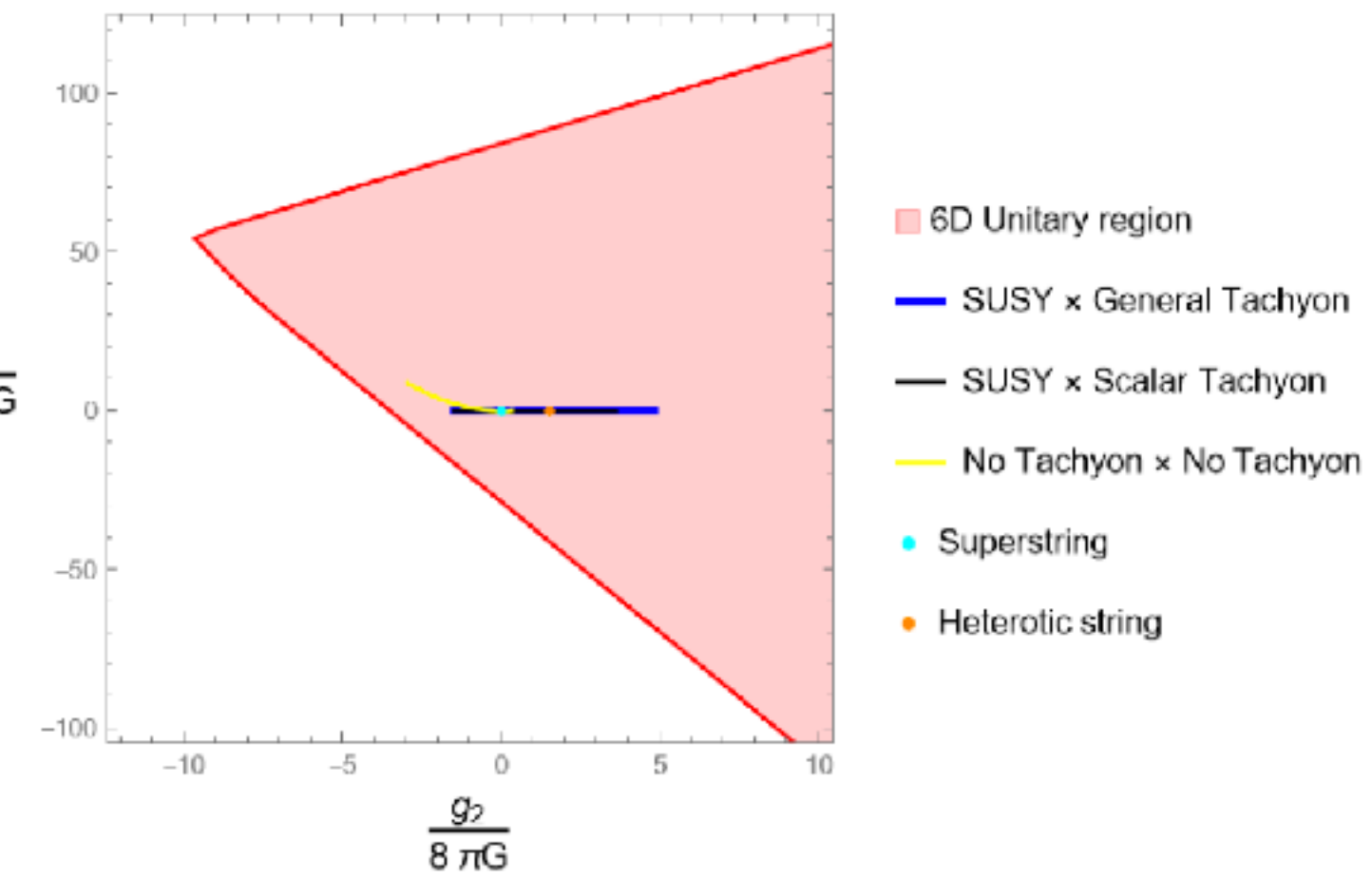
$$\sigma_2 = s^2 + t^2 + u^2, \quad \sigma_3 = stu$$

$$M(s, t) = 8\pi G \left( \frac{tu}{s} + \frac{su}{t} + \frac{st}{u} \right) + \lambda_1 \left( \frac{1}{s} + \frac{1}{t} + \frac{1}{u} \right) + \lambda_2 + g_2 \sigma_2 + g_3 \sigma_3 + g_4 \sigma_2^2 + \dots$$

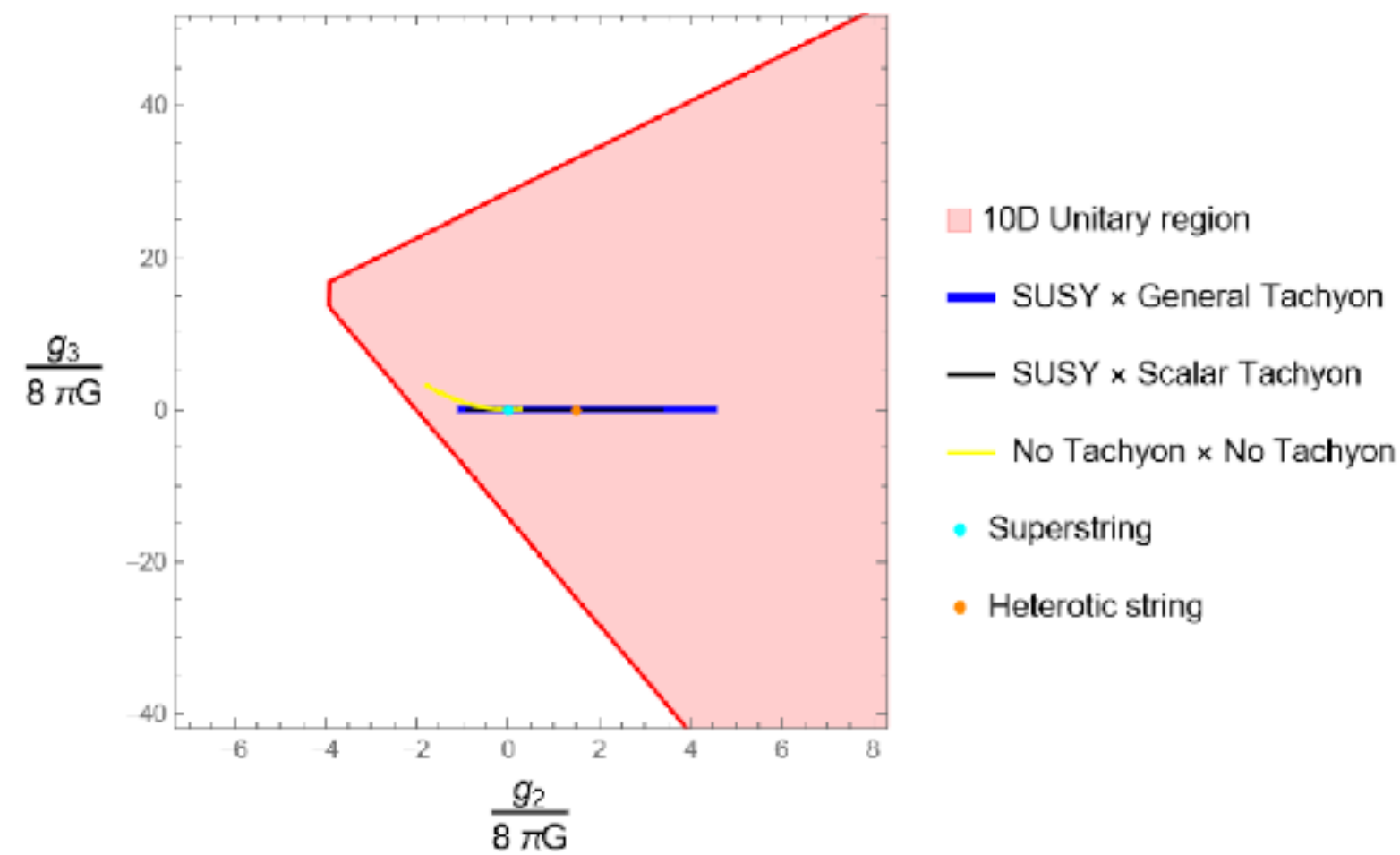
- non-SUSY x non-SUSY:  $\lambda_{1,2} = 0, \quad g_2 = 8\pi G g_{1,0}, \quad g_3 = 8\pi G g_{1,0}^2, \quad g_4 = 4\pi G g_{3,0}$
- SUSY x non-SUSY (tachyonic):  $\lambda_{1,2} = 0, \quad g_2 = 4\pi G g_{1,0}^{Tach}, \quad g_3 = 0, \quad g_4 = 2\pi G (g_{3,0}^{Tach} + g_{1,0}^{SUSY})$



(a) 5D unitary region and the three scenarios of KLT double copy.



(b) 6D unitary region and the three scenarios of KLT double copy.



(c) 10D unitary region and the three scenarios of KLT double copy.

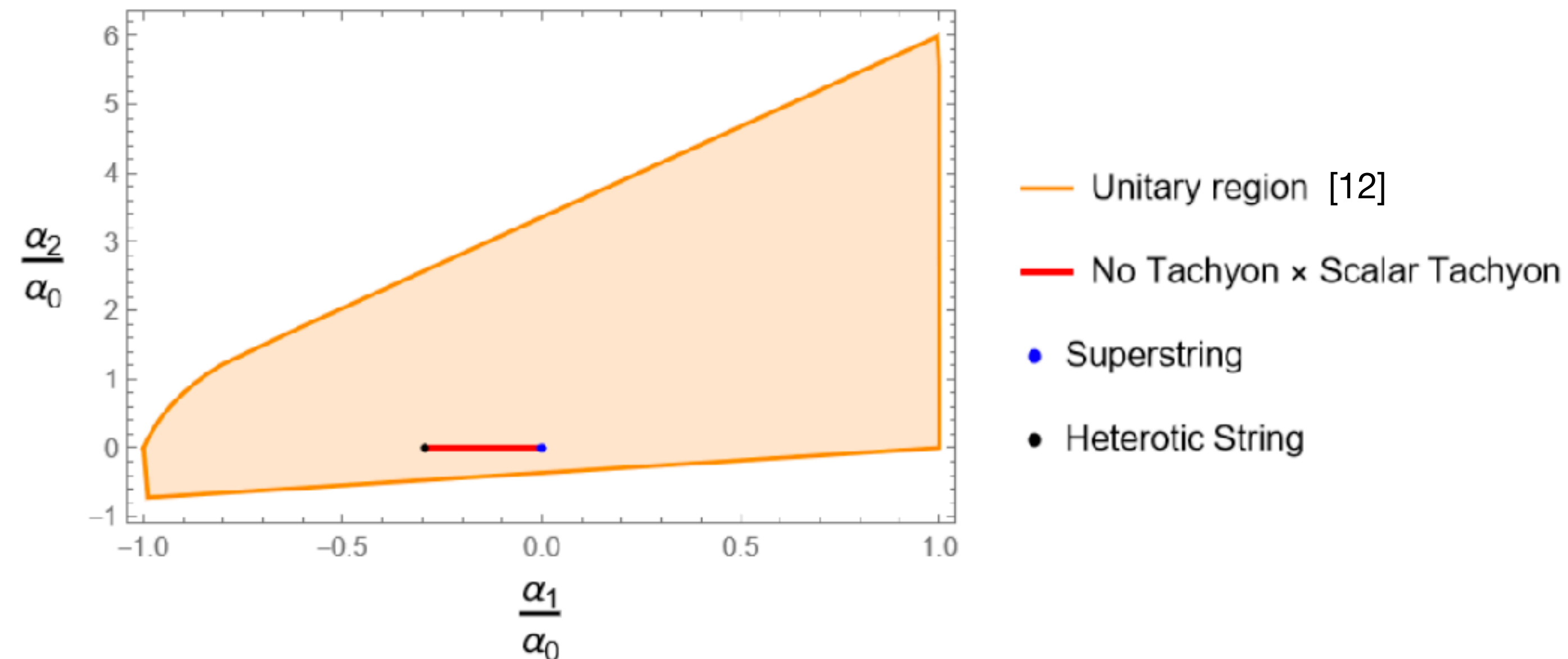
Unitarity of the open string couplings appears to imply that of the closed string.



- Four-graviton amplitude through the double copy

$$M(1^+, 2^+, 3^-, 4^-) = \pi [12]^4 \langle 34 \rangle^4 \left[ \frac{1}{stu} + \gamma_1 \frac{1}{s} + \gamma_2 \frac{t^2}{s} + \alpha_0 + \alpha_1 s + \alpha_2 t \right] = \sin(\pi t) A(1^+, 2^+, 3^-, 4^-) A(1^+, 3^-, 2^+, 4^-)$$

$$\gamma_1 = 2\tilde{c}, \quad \gamma_2 = -\tilde{c}^2, \quad \alpha_0 = 2\tilde{b}_{1,0}, \quad \alpha_1 = 2\tilde{b}_{2,0} - \frac{\pi^4}{45}, \quad \alpha_2 = -\tilde{c}^2.$$





# Summary

- EFThedron + SUSY + string monodromy  $\implies$  a tiny island surrounding the type-I superstring EFT.
- We get two-sided bounds within  $10^{-4}$  of the superstring result and prove that the critical dimension is 10 with only a finite number of low-energy couplings.
- Relaxing SUSY, the allowed region becomes a strip, and the spacetime dimension is at most 12.
- For gluon scattering, the coefficients are again pinned down to the superstring value.
- Introducing tachyons to the bootstrap, the space is spanned by superstring & bosonic string.
- Closed string EFTs are obtained using the double copy. Unitarity of the open string appears to imply that of the closed string.
- Future directions